

PART 1 – CYBER SECURITY (Complete)

Unit 1 ki pehli building block. Cyber Security samajhne ke liye pehle ye janna zaroori hai ki wo kya hoti hai, kisko protect karti hai, aur kyu zaroori hai. Aage ke saare chapters (Cyber Attack, Cyber Crime) isi par khade hain.

1.1 Cyber Security Kya Hoti Hai?

PRACTICAL EXAMPLE – REAL LIFE

Socho tumhare Papa ne ek naya ghar banaya. Us ghar me bahut important cheezein hain: Rs. 5 lakh cash, gold jewellery, laptop, documents (Aadhaar, PAN, Property Papers), TV, mobile, family photos. Ab tumhare Papa bina lock lagaye ghar chhod dete hain. Kya ho sakta hai? Chor andar aa sakta hai, paise chori ho sakte hain, jewellery le ja sakta hai, documents jala sakta hai, ghar ka samaan tod sakta hai. Isliye Papa security lagate hain: strong door, lock, CCTV, alarm, security guard, fingerprint lock. Ab chor ke liye ghar me ghusna bahut mushkil ho gaya.

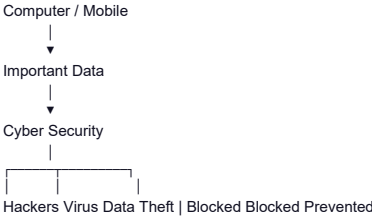
PRACTICAL EXAMPLE – REAL LIFE

Aaj har insaan ka ek Digital Ghar hota hai. Usme kya hota hai? Gmail, WhatsApp, Instagram, Facebook, College Portal, Bank Account, UPI, ATM Details, Aadhaar, PAN, Photos, Passwords, Company Files. Ab socho agar koi hacker is digital ghar me ghus jaye to wo kya kar sakta hai? Password chori, bank account se paise transfer, photos leak, Gmail hack, Instagram account delete, college marks change, company ka data chori. Bilkul waise hi jaise chor tumhare ghar me ghus gaya tha. Is digital ghar ko bachane ke liye jo security lagayi jaati hai, usse Cyber Security kehte hain.

DEFINITION

Cyber Security woh process hai jo computers, mobiles, servers, networks aur data ko hackers, viruses aur unauthorized access se protect karti hai.

FLOWCHART



1.2 Cyber Security Kiski Security Karti Hai?

Cyber Security sirf computer ki security nahi karti. Ye protect karti hai:

- Computer, Mobile, Network, Cloud, Database, Server
- Email, Online Banking, Online Shopping
- Hospital Records, College Data, Company Information

1.3 Cyber Security Ki Zarurat Kyu Hai?

Agar Cyber Security na ho to:

- Password chori ho sakta hai.
- ATM account hack ho sakta hai, WhatsApp hack ho sakta hai.
- Company ka data leak ho sakta hai.
- Hospital records delete ho sakte hain.
- Government websites hack ho sakti hain.

Daily Life Examples: ATM PIN, Phone Lock, Face Unlock, OTP, Gmail Password, Antivirus, Firewall — ye sab choti-choti security hi to hain jo hamari digital life protect karti hain.

1.4 Cyber Security Ke Objectives (Goals)

- Data ko safe rakhna — jaise bank balance.
- Privacy bachana — jaise WhatsApp chats.
- Unauthorized access rokna — sirf owner hi account khol sake.
- Virus rokna — computer ko damage hone se bachana.
- Online fraud rokna — fake websites aur scams se protection.

1.5 Cyber Security Ke Main Components

FLOWCHART



1.6 Cyber Security Kahan Use Hoti Hai?

Jagah | Kyu | Bank | Money protect karne ke liye | Hospital | Patient records protect karne ke liye | College | Student data aur marks protect karne ke liye | Amazon / Flipkart | Customer payment secure karne ke liye | Airport | Flight systems ko secure rakhne ke liye

Army Secret military information protect karne ke liye

1.7 Cyber Security Ke Benefits

- Data Safe, Privacy Safe, Money Safe, Identity Safe
- Business Safe, Trust Increase
- Online Transactions Safe

1.8 Agar Cyber Security Na Ho To?

Imagine ek din: tumhara Gmail hack, WhatsApp delete, Instagram kisi aur ke control me, bank account se Rs. 50,000 transfer, college portal me marks change, laptop ki files encrypt. Ye sab ek hi din me ho sakta hai — isi liye Cyber Security bahut important hai.

EXAM NOTES (2 MARKS / VIVA)

Exam Definition (2 Marks): Cyber Security is the practice of protecting computers, networks, devices and data from unauthorized access, cyber attacks and digital threats.

Viva Q1. Cyber Security kya protect karti hai? Ans: Computer, Network, Server, Mobile aur Data.

Viva Q2. Cyber Security ka main purpose kya hai? Ans: Data ko unauthorized access aur attacks se protect karna.

MEMORY TRICK

Cyber Security = Digital Ghar ki Security. Jaise ghar ko lock, CCTV aur guard protect karte hain, waise hi Cyber Security hamare digital data ko hackers aur attacks se protect karti hai.

PART 2 – CYBER ATTACK (Complete)

Foundation chapter. Cyber Attack ka concept samajhna zaroori hai kyunki Cyber Crime (Unit 1) aur Active/Passive Attack dono isi ki branches hain. Dhyan se padho — ye sabki root hai.

2.1 Cyber Attack Kya Hota Hai?

PRACTICAL EXAMPLE – REAL LIFE

Socho tumhare school me ek Computer Lab hai jisme 40 computers hain. Usme save hai: students ki marksheet, fees records, teachers ki salary, exam papers, principal ke documents, CCTV recordings. Ab maan lo ek din ek hacker internet ke through school ke server se connect ho gaya. Wo kya kar sakta hai? Situation 1: Sabhi students ke marks change kar deta hai — 90 → 40, 40 → 95. Situation 2: Principal ka password change kar deta hai — ab Principal khud login nahi kar pa rahe. Situation 3: Exam paper internet par leak kar deta hai. Situation 4: Saari files delete kar deta hai. Situation 5: School ki website band kar deta hai. Situation 6: Students ki personal information (mobile, Aadhaar, address, parents details) chori kar leta hai. Socho, school ko kitna bada loss hoga. Ye jo hacker ne jaan-boojhkar system ko nuksan pahunchaya, ise Cyber Attack kehte hain.

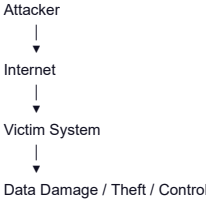
PRACTICAL EXAMPLE – REAL LIFE

Tumhara Gmail password hai Vinay@123. Ek hacker ne kisi tarah password jaan liya. Fir usne Gmail login kiya, password change kar diya, recovery email badal di aur OTP number bhi badal diya. Ab Gmail tumhara tha, lekin control hacker ke paas chala gaya. Ye bhi Cyber Attack hai.

DEFINITION

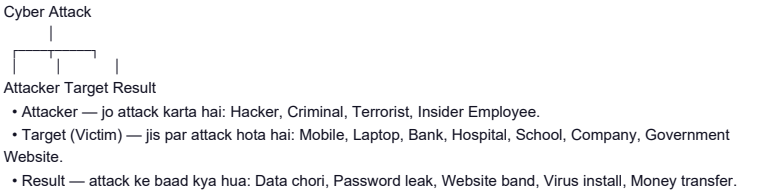
Cyber Attack ek intentional (jaan-boojhkar) attempt hota hai jisme attacker computer, network, server ya data ko access, steal, damage, modify ya destroy karne ki koshish karta hai.

FLOWCHART



2.2 Cyber Attack Me 3 Main Characters Hote Hain

DIAGRAM



2.3 Cyber Attack Kyu Kiya Jata Hai? (Motives)

Bahut log sochte hain hacker sirf maze ke liye hack karta hai. Reality me reasons alag hote hain:

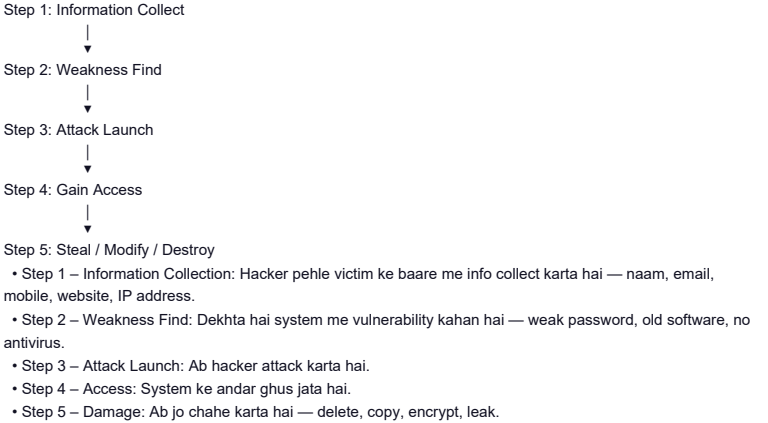
Motives | Example

1. Money Bank account hack, UPI fraud, Credit Card fraud (sabse common)
2. Data Chori | Company ke secret documents chori karna | 3. Identity Chori | Tumhare naam se fake account banana
4. Revenge Ex-employee company ka database delete kar deta hai
5. Fame Kuch hackers famous hone ke liye website hack karte hain
6. Political Reason Ek country dusri country ke systems par attack karti hai
7. Terrorism Power grid, airport ya railway system ko target karna

2.4 Cyber Attack Ka Process (Attack Lifecycle)

Har attack ek hi second me nahi hota. Generally ye steps hote hain:

FLOWCHART

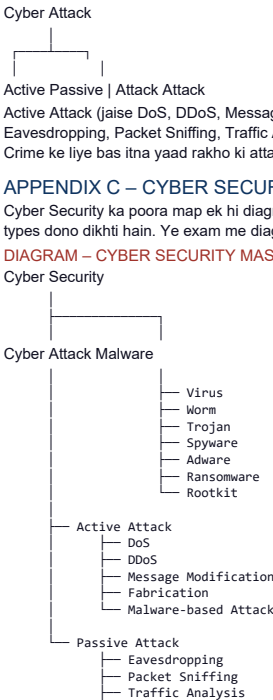


2.5 Cyber Attack Kin Cheezon Par Ho Sakta Hai?

Computer, Mobile, Wi-Fi, Server, Website, Email, Cloud, ATM, Smart TV, CCTV, IoT Devices — koi bhi digital device target ban sakta hai.

2.6 Cyber Attack Ke Main Categories

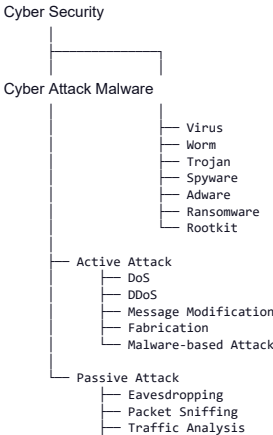
FLOWCHART



APPENDIX C – CYBER SECURITY MASTER TREE

Cyber Security ka poora map ek hi diagram me — isme Cyber Attack ki types (Active/Passive) aur Malware ki types dono dikhti hain. Ye exam me diagram ke liye ekdum ready hai.

DIAGRAM – CYBER SECURITY MASTER TREE



Wiretapping

Active Attack = jo system me badlaav karta hai (modify/destroy) — jaise DoS, DDoS, Message Modification, Fabrication, Malware-based Attack. Passive Attack = jo sirf dekh-ta hai (data churakar padhta hai, badlaav nahi karta) — jaise Eavesdropping, Packet Sniffing, Traffic Analysis, Wiretapping.

Attack | Type | Kya hota hai | DoS / DDoS | Active | System par overload karke service band karna | Message Modification | Active | Message ke andar data badal dena | Fabrication | Active | Jhootha / fake message ghussa dena | Malware-based Attack | Active | Virus/ransomware install kar ke nuksan | Eavesdropping | Passive | Chupke se baat-chit sunna | Packet Sniffing | Passive | Network traffic ke packets padhna

Traffic Analysis Passive Data ko padhe bina uske patterns se info nikalna

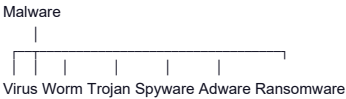
Wiretapping | Passive | Kisi aur ki line/communication tap karna

APPENDIX B – MALWARE TREE

Malware = Malicious software — wo programs jo computer ko nuksan pahunchate hain. Ye exact tree exam ke

diagram sawal ke liye ready hai.

DIAGRAM – MALWARE CLASSIFICATION TREE



Rootkit | Keylogger | Bot | Botnet

Ye tree exam me 3 tarah se poochha jata hai: (1) Malware kya hai — definition, (2) Malware ki types — tree

diagram, (3) kisi ek type ko explain karo — table se.

Malware | Kya karta hai | Example

Virus Files me chhip kar phailta hai, damage karta hai ILOVEYOU

Worm Bina kisi file ke khud copy hoke network par phailta hai Morris Worm

Trojan | Achha software ban ke andar ghus jata hai | Banking trojans | Spyware | Bina bataye aapki activity dekhta hai | Keyloggers, trackers | Adware | Zabardasti ads dikhta hai | Pop-up ads | Ransomware | Data lock (encrypt) karke paisa maangta hai | WannaCry

Rootkit System ki jad me chhip jata hai, detection se bachta hai Hiding in kernel

Keylogger Har key press record karta hai (password churata hai) Bank password steal

Bot / Botnet | Aapka system hacker ke remote control me | Spam-sending botnets

APPENDIX A – SECURITY SERVICES (CIA TRIAD)

Cyber Security ka poora foundation CIA Triad par khada hai. Exam me ye sabse poocha jaane wala diagram hai. Pehle exact diagram, phir explanation.

DIAGRAM – CIA TRIAD

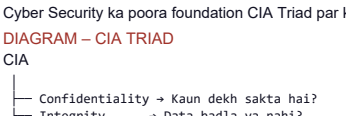


DIAGRAM – BAQI SERVICES

Baaki Security Services:

Authentication → Tum kaun ho? Authorization → Tum kya kar sakte ho? Accountability → Kisne kya kiya? Non-Repudiation → Kiya hai to mana nahi kar sakte. Accessibility → Authorized user aasani se use kar sake.

Privacy → Personal information ko protect karna.

DEFINITION

Confidentiality: Data ko sirf authorized (allowed) log dekh saken. Integrity: Data ko koi badal / corrupt na kar sake. Availability: Authorized user ko zarurat par data hamesha milta rahe.

Service | Sawaal | Matlab | Attack Jo Todta Hai

Confidentiality Kaun dekh sakta hai? Data sirf authorized log hi dekh saken Eavesdropping, Sniffing

Integrity | Data badla ya nahi? | Data original aur galat nahi hona chahiye | Message Modification | Availability | Zarurat par mil raha hai? | Authorized user ko kabhi bhi data mile | DoS / DDoS | Service | Sawaal | Matlab

Authentication Tum kaun ho? Pehchan verify karna (password, OTP, fingerprint)

Authorization | Tum kya kar sakte ho? | Access level decide karna (kya allowed hai) | Accountability | Kisne kya kiya? | Har kaam ka record hona chahiye

Non-Repudiation Kiya hai to mana nahi kar sakteKiya hua kaam deny nahi kar sakte (digital signature)

Accessibility | Aasani se use kar sake | System sirf aapke liye kaam kare | Privacy | Personal info protect | Personal data kisi aur ke haath na jaye

2.7 Cyber Attack Ke Possible Results

Result | Example | Data Theft | Password chori | Data Modification | Marks badal dena | Data Deletion | Files delete | Financial Loss | Bank fraud | Service Down | Website crash | Privacy Loss | Personal photos leak

2.8 Cyber Attack Ko Rokne Ke Tarike (Prevention)

- Strong Password, 2-Factor Authentication (2FA)
- Antivirus, Firewall, Software Update, Backup
- Unknown links par click na karna, Public Wi-Fi par sensitive kaam na karna

EXAM NOTES (2 MARKS / VIVA)

Exam Definition (2 Marks): Cyber Attack is an intentional attempt to gain unauthorized access to a computer system, network, or data in order to steal, modify, destroy, or disrupt information or services.

Viva Q1. Cyber Attack kya hai? Ans: Kisi computer ya network par unauthorized access ya damage pahunchane ki koshish ko Cyber Attack kehte hain.

Viva Q2. Cyber Attack ka sabse common objective kya hai? Ans: Data theft, financial gain, aur system ko damage karna.

MEMORY TRICK

Cyber Attack = Digital Chor. Jaise ghar me chor ghuskar paisa ya samaan chori karta hai, waise hi cyber attack me attacker digital system me ghuskar data, password ya paise chori karta hai ya system ko nuksan pahunchata hai.

PART 3 – INTRODUCTION OF CYBER CRIME

Unit 1 ka asli syllabus yahan se shuru hota hai. Is chapter me hum samjhenge ki Cyber Crime kya hai, iske elements kya hain, ye traditional crime se kaise alag hai, aur kis kis category me divide hota hai.

3.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Meera Delhi me ek bank me kaam karti hai. Ek din uske phone par WhatsApp message aaya: "Congratulations! Aapne Rs. 5,00,000 ki lottery jeeti hai. Claim karne ke liye is link par click karein." Meera ne link click kiya. Ek page khula jisme likha tha — "Processing fee Rs. 2,000 bharo". Usne UPI se bhar diye. Phir aise hi do aur fees bhari. Total Rs. 12,000 chale gaye. Lottery ka paisa kabhi nahi aaya. Ye ek Cyber Crime hai (Online Fraud).

Usi hafte aur bhi kuch hua:

- Meera ke friend Rohit ke Instagram account se uske followers ko "free iPhone" messages gaye — account hack ho gaya.
- College portal me kisi ne marks change kar diye.
- Ek kaam-wali didi ke Aadhaar se fake SIM nikal li gayi.

Ye saare incidents ek cheez me same hain — computer / mobile / internet ka istemal karke criminal kaam hua. Isi ko hum Cyber Crime kehte hain.

3.2 Definition

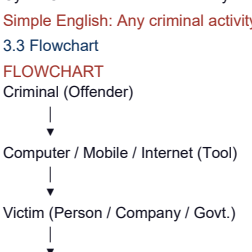
DEFINITION

Cyber Crime woh criminal activity hai jisme computer, mobile ya internet ka istemal karke koi illegal kaam kiya jata hai — jaise paisa chori karna, data chori karna, kisi ko dhamkana, ya kisi system ko nuksan pahunchana.

Simple English: Any criminal activity that involves a computer or the internet is called Cyber Crime.

3.3 Flowchart

FLOWCHART



3.4 Elements of Cyber Crime (5 Cheezein Zaroori)

Kisi bhi cyber crime me ye 5 elements hote hain — agar ek bhi missing ho to cyber crime nahi banega:

Element | Kya hai | Example | 1. Offender | Jo galat kaam karta hai | Hacker, fraud gang, insider employee | 2. Tool | Jis se galat kaam hota hai | Computer, internet, mobile, software | 3. Victim | Jis par galat kaam hota hai | Person, company, bank, government | 4. Act | Illegal activity khud | Hacking, fraud, spoofing, spam | 5. Loss / Impact | Hone wala nuksan | Money, data, privacy, reputation

3.5 Cyber Crime vs Traditional Crime

Cyber Crime ko samajhne ka sabse aasan tarika hai ise normal (traditional) crime se compare karna:

Basis | Traditional Crime | Cyber Crime | Jagah | Sirf ek jagah (local) | Internet hai to poore world me | Tool | Physical (chaku, bandook) | Computer, internet, software | Criminal ki pehchan | Pata chalti hai (face, CCTV) | Chhupi hoti hai (fake ID, VPN)

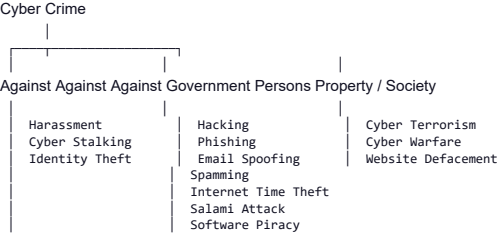
Evidence Physical (fingerprint, weapon) Digital (logs, IP, files) — delete ho sakta hai

Speed | Dheere hota hai | Seconds me khatam | Distance | Criminal ko paas hona zaroori | Duniya ke kisi bhi kone se | Jurisdiction | Ek hi desh/court ka law | Kaunsa desh ka law lagega — confusion | Record | Bahut kam records | Sab kuch digital record rehta hai

3.6 Classifications of Cyber Crimes (Overview)

Cyber Crimes ko target ke hisaab se 3 bade categories me divide kiya jata hai:

FLOWCHART



IMPORTANT POINTS

Unit 1 syllabus me 4 classifications detail me padhenge: Email Spoofing, Spamming, Internet Time Theft, Salami Attack — ye sab 'Against Property' category ke cyber crimes hain. Inhe Part 5 me poore detail me padha jayega.

3.7 Characteristics of Cyber Crime

- Anonymity — criminal apni pehchan chhupa kar kaam karta hai.
- Global reach — duniya ke kisi bhi country ko target kar sakta hai.
- Low cost — bas ek computer + internet chahiye.
- No boundaries — borders nahi maante.
- Instant — seconds me kaam ho jata hai.
- Hard to detect — evidence delete karna aasan hai.
- High impact — ek attack se lakhon logon ko nuksan ho sakta hai.

3.8 Who Commits Cyber Crime? (Types of Cyber Criminals)

Criminal Type | Kaun hai | Kya karta hai | Hackers | White / Black / Grey hat | System me ghusne wale technical log | Organized Criminals | Gangs | Bank fraud, card fraud, extortion | Insiders | Company ke andar ka employee | Andar ki information leak karta hai | Terrorists | Desh ke khilaf groups | Cyber terrorism, critical systems par attack | Disgruntled Employees | Ex-employee | Badla lene ke liye data delete/leak | Script Kiddies | Naye log | Ready-made tools use karke khelne wale

3.9 Real World Examples (Chote Chote)

- WannaCry Ransomware (2017) — duniya bhar ke hospitals aur laptops band kar diye.
- Bank SIM-swap frauds in India — SIM block karke OTP chura kar paise nikalna.
- Aadhaar data leaks — personal data market me bikna.
- Deepfake videos — kisi ki fake video banakar blackmail.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes – One Line Definitions:

1. Cyber Crime = computer/internet ke through ki gayi criminal activity. 2. Cyber Crime ke 5 elements = Offender, Tool, Victim, Act, Loss. 3. Cyber Crime ki 3 categories = Against Person, Against Property, Against Government. 4. Cyber crime me criminal chhupa rehta hai, par digital evidence rehta hai.

Viva Q1. Cyber Crime me 'tool' kya hota hai? Ans: Computer, mobile, ya internet jiske through crime hota hai.

Viva Q2. Cyber crime ko traditional crime se alag kya banata hai? Ans: Anonymity, global reach, speed aur digital evidence — criminal ko victim ke paas hone ki zaroorat nahi.

MEMORY TRICK

Cyber Crime = Digital Daku. Jaise highway par daku gaadi rok kar lootta hai, waise hi internet par cyber criminal system rok kar data aur paise lootta hai — bas farak itna hai ki daku ko face dikhata hai, cyber criminal nahi.

PART 4 – CHALLENGES OF CYBER CRIME

Crime hota hai, lekin usse pakadna itna aasan nahi hota. Is chapter me hum wo saari problems padhenge jo cyber crimes ko rokne, pakadne aur saza dilane me aati hain.

4.1 Long Practical Example (Real Life)

PRACTICAL EXAMPLE – REAL LIFE

Ravi ke bank account se Rs. 2,00,000 gayab ho gaye. Police ne case khola. Lekin investigation me har step par rukawat aayi: 1. Account se paisa 3 alag accounts me gaya, phir 50 aur accounts me transfer hua. 2. Sab transactions VPN se hui — IP address fake tha. 3. Attacker South Africa me tha, server America me, victim India me. 4. Paisa cryptocurrency (Bitcoin) me badal diya gaya — track karna near impossible. 5. Jab tak police ko pata chala, 6 mahine nikal chuke the — digital records delete ho chuke the.

Police ke saamne problem: criminal pakadna mushkil, saza dilana aur bhi mushkil. Yehi problems — jo cyber crime se nipatne me aati hain — Challenges of Cyber Crime kehlati hain.

4.2 Definition

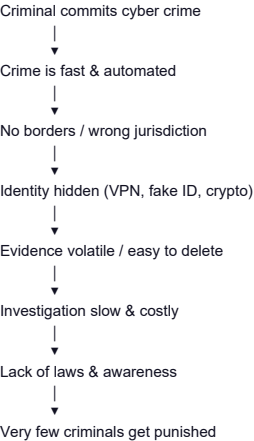
DEFINITION

Challenges of Cyber Crime wo difficulties / problems hain jo cyber crimes ko prevent (rokne), detect (pakadne), investigate (jaanch) aur punish (saza dilane) me saamne aati hain.

Simple: Crime karna aasan, lekin cyber crime ko pakadna — kyunki criminal chhupa hua hai, country ke bahar hai, aur evidence mit jata hai.

4.3 Flowchart – Challenges Ki Chain

FLOWCHART



4.4 Detailed Challenges (Sabhi Point Detail Me)

1. Anonymity (Pehchan Chhupana)

Attacker VPN, Tor browser, fake accounts aur stolen SIM istemal karta hai jisse uski asli pehchan kabhi saamne nahi aati. Example: fraud karne ke baad account fake naam se khula milta hai, asli banda kisi aur country me hota hai.

2. Jurisdiction Problem (Kaunsi Police Case Kare)

Crime India me hua, server USA me hai, attacker Russia me baitha hai. Ab kaunsi country ki police case kare? Har desh ke laws alag hote hain, aur cross-border cooperation me mahino lag jaate hain.

3. Global / Borderless Nature

Internet par koi border nahi hota. Ek click me attacker duniya ke kisi bhi system par attack kar sakta hai. Example: ek country ke hacker dusre country ke bank par hamlā — bina wahan gaye.

4. Evidence Collection (Saboot Jutana Mushkil)

Digital evidence (logs, cookies, IP records, files) bahut nazuk hota hai — computer band karo ya system restart karo to data jaa sakta hai. Delete karna bhi ek click ka kaam hai.

5. Fast & Automated Attacks

Bots aur scripts ek second me lakhon attacks kar dete hain. Police/security ke paas itna time hi nahi hota ki wo har attack ko rok saken.

6. Encryption (Data Lock)

Attacker apna data encrypt kar leta hai (jise bina key ke koi nahi padh sakta). Police ke liye encrypted data ko padhna aur prove karna bahut mushkil ho jata hai.

7. Rapidly Changing Technology

Roz nayi technology aur naye tareeke ke crimes bante hain. Laws aur police piche reh jaate hain kyunki naya crime samajhne me time lagta hai.

8. Lack of Awareness (Sabse Bada Weak Point)

Zyaada tar log OTP share kar dete hain, unknown links par click karte hain, weak passwords rakhte hain. Insaan hi sabse bada weak link hai. Cyber security itni strong nahi hoti ki user ko bacha sake.

9. Lack of Technical Skills in Investigation

Aam police ko cyber ka training nahi hoti. Cyber cell / forensic lab mahangi aur kam hoti hai, isliye investigation slow aur incomplete rehti hai.

10. Cross-border Legal Differences

Kuch desh me cyber crime ke strict laws hi nahi hain — aise desh cyber criminals ke liye safe haven ban jaate hain jahan se wo bina dar ke kaam karte hain.

11. High Cost of Security

Strong security (firewall, SIEM, cyber insurance, trained staff) bahut mahangi hai. Chhoti companies aur schools afford nahi kar paati — isliye wo aasan target ban jati hain.

12. Difficult Prosecution (Saza Dilana Mushkil)

Sahi saboot, sahi jurisdiction aur international cooperation ke bina court me conviction rate bahut kam hota hai. Criminal pakda bhi jaye to punish karna mushkil hai.

4.5 Diagram – Challenge vs Reason vs Solution

Exam me likhne ke liye ye compact table sabse useful hai:

Challenge | Problem Kyu Hai | Kya Solution Ho Sakta Hai | Anonymity | Criminal ki pehchan chhupi rehti hai | Digital footprint tracking, cyber forensics

Jurisdiction Crime ka desh clear nahi International treaties, mutual legal assistance

Borderless nature | Koi border nahi | Global cyber security cooperation | Evidence | Digital data volatile / deletable | Forensic imaging, proper chain of custody | Fast attacks | Seconds me ho jate hain | Automated detection, AI monitoring | Encryption | Data lock ho jata hai | Legal decryption procedures, key escrow | Technology change | Laws piche reh jate hain | Regular law updates | Lack of awareness | User khud dhoka kha jata hai | Cyber awareness campaigns, education | Low conviction | Proof + jurisdiction mushkil | Stronger laws, cyber courts

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Challenge = cyber crime ko rokne / pakadne me aane wali problem. 2. Sabse bada challenge = Jurisdiction + Anonymity (crime ka desh aur criminal ki pehchan). 3. Evidence digital hota hai — mit sakta hai, isliye jaldi collect karna zaroori.

Viva Q1. Cyber crime ko pakadna itna mushkil kyu hai? Ans: Criminal chhupa hota hai (VPN/fake ID), crime country se bahar hota hai (jurisdiction), aur digital evidence mit jata hai.

Viva Q2. Insaan sabse bada weak point kyu? Ans: Kyonki user OTP/share karta hai, unknown links par click karta hai aur weak passwords rakhta hai.

MEMORY TRICK

Challenge = Chhup-kar kaam karne wala criminal. Jaise chor raat ke andhere me chhup jata hai, waise cyber criminal VPN ke andhere me chhup jata hai — pakadna mushkil isliye kyunki koi face hi nahi dikhta.

PART 5 – CLASSIFICATIONS OF CYBER CRIMES

Unit 1 ka sabse important chapter. Isme hum syllabus ke 4 classifications ko ek-ek karke detail me padhenge: Email Spoofing, Spamming, Internet Time Theft, Salami Attack. Har ek me pattern same rahega — Example →

Definition → Diagram/Flowchart → Prevention → Exam Notes.

5.0 Overview – Cyber Crimes Ki Classification

FLOWCHART

Classifications of Cyber Crimes



Email Spamming Internet Salami Spoofing Time Theft Attack

Iske alawa bhi kai classifications hain (Phishing, Hacking, Cyber Stalking, Software Piracy, etc.) — lekin syllabus me yahi 4 main hain, aur inhi ko exam me detail me likhna hai.

5.1 EMAIL SPOOFING

PRACTICAL EXAMPLE – REAL LIFE

Riya ko ek email mila jo bilkul uske bank se dikh raha tha — sender: support@sbibank.com. Email me likha tha: "Aapka account block hone wala hai. Verify karne ke liye is link par click karein." Riya ne link par click kiya — ek page khula jo bank ke page jaisa hi tha. Usne apni user ID, password aur OTP daal diye. Agle hi minute uske account se Rs. 40,000 nikal gaye.

Sach: wo email bank ne nahi bheja tha. Sender ka address fake tha — attacker ne bank jaisa dikhne ke liye bana diya tha. Isi ko Email Spoofing kehte hain.

DEFINITION

Email Spoofing ek cyber crime hai jisme attacker email ke sender address ko fake karke aisa dikhata hai jaise email kisi trusted person / company se aaya ho — taaki victim dhoka kha kar apni sensitive information (password, OTP, bank details) de de ya malware install kar le.

Simple: Jisne bheja hai uska naam jhootha likhna, taaki tumhe lagge ki trusted logon ne bheja hai.

Kaise Kaam Karta Hai (Diagram)

DIAGRAM – HOW EMAIL SPOOFING WORKS



Password / OTP chori OR Malware install

IMPORTANT POINTS

Ye kyun possible hai? Email bhejne wala protocol SMTP (Simple Mail Transfer Protocol) sender ka address verify nahi karta. Jaise koi letter par bina pakke hue jhootha naam likh de — post office rokta nahi. Waise hi email system bhi sender ko check nahi karta.

Types of Email Spoofing

Type | Kya hota hai | Example

1. Display Name Spoofing Sirf naam trusted dikhata hai, email ID alag hoti hai Riya Verma <riya123@gmail.com>
2. Legitimate Domain Spoofing Same company ka domain hack karke use karta hai support@sbibank.com (hacked)
3. Look-alike Domain Spoofing | Similar dikhne wala domain | sbibank.com vs sbibank-secure.com | 4. Reply-to Spoofing | Reply jab karo to attacker ke paas jata hai | Reply-to: fraud@fake.com

Email Spoofing vs Phishing

Spoofing = sender address fake karna (technique). Phishing = dhoka dena (purpose). Spoofing ka istemal phishing karne ke liye hota hai. Matlab — spoofing chabi hai, phishing taala hai jo wo khulta hai. Dono aksar saath saath aate hain.

Prevention (Bachao Ke Tarike)

- Sender ka email address dhyan se dekho (domain check karo).
- Link par hover karke URL check karo — click karne se pehle.
- Urgent / dhamki wale messages par suspicious raho.
- OTP kabhi kisi ko na do — bank bhi OTP nahi maangta.
- Company/bank par call karke verify karo.
- Organizations apne email par SPF, DKIM, DMARC records enable karein.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Email Spoofing = sender ka address fake karna. 2. Target = trust jeet kar password/OTP churana. 3. Ye isliye possible hai kyunki SMTP sender ko verify nahi karta.

Viva Q1. Email spoofing me kya fake hota hai? | Ans: Sender ka address / identity.

Viva Q2. Spoofing aur phishing me kya relation hai? Ans: Spoofing ek technique hai jo phishing (dhoka) karne ke liye use hoti hai.

5.2 SPAMMING

PRACTICAL EXAMPLE – REAL LIFE

Aman ka mobile har din 30 se zyada spam messages se bhara rehta hai:

- "Aapko sirf Rs. 99 me smartphone milega!"
- "Congratulations! Prize jitne ke liye reply karein"

- Random loan / gambling / casino ads

Aur uske Gmail me bhi 100 unread promotional emails — asli emails dhoondna mushkil ho jata hai. Agar Aman inme se kisi link par click kar de — malware ya fraud. Yehi Spamming hai.

DEFINITION

Spamming ek cyber crime hai jisme kisi ko bina uski permission ke bulk (bahut zyada) unwanted messages, emails ya advertisements bheje jaate hain.

Simple: Kisi aur ka inbox/mobile bina pooche-jhadhe ads aur bekaar messages se bhar dena.

FLOWCHART



Types of Spam

Type | Kya hai | Email Spam | Bulk promotional / junk emails | SMS Spam | Bulk messages on mobile

Comment Spam Blogs / forums par fake comments (link ke saath)

Search Spam | Search results me fake pages dikhana | Social Media Spam | Fake accounts se bulk messages / tags

Spamming Ka Nuksan (Harm)

- Time waste — real messages dhoondne me time lagta hai.
- Bandwidth / data khatam — mobile data aur internet bill badta hai.
- Real email miss ho jate hain (important mail spam me dab jati hai).
- Phishing aur malware ka zariya — spam ke andar harmful links hote hain.
- Fraud ka gate — spam me fake offers lottery scams bheje jate hain.

Spam vs Scam

Spam = koi bhi unwanted bulk message. Scam = us message ke andar ka dhoka (paisa chori ki koshish). Spam scam ka carrier ban sakta hai — spam message ke andar scam ka link hota hai.

Prevention (Bachao Ke Tarike)

- Spam filter ON rakho (email apps me default hota hai).
- Apna email / phone number public jagah share na karo (social media, random sites).
- Trusted sites par hi unsubscribe karo.
- Unknown links par click mat karo — reply bhi mat karo.
- Spam ko report karo.
- Sign-ups ke liye alag email ID banao (junk email).

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Spamming = bina permission ke bulk unwanted messages bhejna. 2. Spam ka sabse bada nuksan = time/bandwidth waste + phishing ka zariya. 3. Spam vs Scam = spam carrier hai, scam dhoka hai.

Viva Q1. Spamming ko cyber crime kyu mana jata hai? Ans: Kyunki ye bina permission ke doosron ke inbox/data par bojh daalta hai aur fraud ka zariya banta hai.

Viva Q2. Spam se bachne ka best tarika? Ans: Spam filter, email public na karna, unknown links par click na karna.

5.3 INTERNET TIME THEFT

PRACTICAL EXAMPLE – REAL LIFE

Ek company har month internet bill deti hai — Rs. 50,000. Employees ka kaam 6 ghante ka hai. Lekin ek employee Suresh roz 3 ghante office internet par online movies dekhta hai, games download karta hai, file sharing karta hai. Iska asar:

- Office ka internet slow ho jata hai — baaki employees ka kaam rukta hai.
 - Company ka bandwidth aur paisa bekaar jaata hai.
 - Company ke data par risk aata hai (pirated content download).
- Yahan company ka paid internet time aur bandwidth bina permission ke personal kaam me use hua — Internet Time Theft.

PRACTICAL EXAMPLE – REAL LIFE

Kabir ke neighbour ne Kabir ka Wi-Fi password guess kar liya. Wo roz Kabir ke internet se downloads karta hai. Ab:

- Kabir ka internet slow ho jata hai.
- Data / FUP khatam ho jata hai, bill badhta hai.

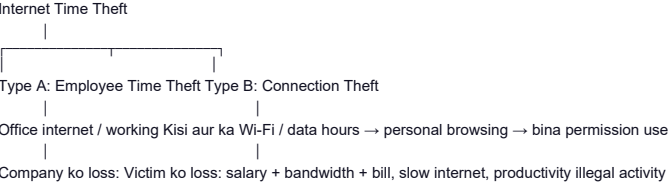
Sabse khatarnak baat — agar neighbour ne internet par koi galat kaam kiya (illegal download, fake post), to IP address ke hisaab se police Kabir ke ghar aayegi kyunki login Kabir ke connection se hua tha. Kabir ko apni bezati ke saath apni bezosai sabit karni padegi.

DEFINITION

Internet Time Theft ek cyber crime hai jisme koi person kisi doosre ki paid internet connection, bandwidth, ya kaam ka samay bina permission ke apne personal / illegal kaam me use karta hai.

Simple: Doosre ka internet / doosre ka kaam ka time — bina pooche apne kaam me lagaana.

FLOWCHART



Examples

- Office me duty hours par online gaming / movie dekhtna.
- Neighbour ka Wi-Fi password hack / guess karke use karna.
- Kisi ka SIM data balance bina permission use karna.
- Shared connection (network spoofing) bina permission use karna.

Prevention (Bachao Ke Tarike)

- Strong Wi-Fi password + WPA2/WPA3 encryption.
- Company me internet usage policy aur monitoring.
- MAC address filtering — sirf known devices hi connect hon.
- Guest network — visitors ke liye alag network.
- Default password turant change karo, time-to-time password change karo.
- Office kaam aur personal kaam ke liye alag connections.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Internet Time Theft = doosre ka paid internet/bandwidth/time bina permission use karna. 2. Do types = Employee Time Theft aur Connection Theft. 3. Danger = illegal activity ka IP blame victim par aata hai.

Viva Q1. Internet Time Theft me kaun kaun loss me hota hai? Ans: Company (salary/bandwidth/productivity) ya victim (bill/slow internet/blame).

Viva Q2. Isse bachne ke liye sabse zaroori kya hai? Ans: Strong password + WPA2/WPA3 aur usage monitoring.

5.4 SALAMI ATTACK / SALAMI TECHNIQUE

PRACTICAL EXAMPLE – REAL LIFE

Ek bank me 10 lakh logon ke accounts hain. Har account par interest credit hota hai. Interest nikalte waqt exact value aati hai, jaise: Rs. 100.0047, Rs. 55.2358 ... Bank aam taur par paise tak round karta hai: Rs. 100.0047 → Rs. 100.00 credit.

Attacker (employee ya hacker) system me chhota sa code daal deta hai jo round karne ke baad bacha fraction (0.0047, 0.2358) alag account me transfer kar deta hai.

- 1 account se kuch nahi — sirf Rs. 0.0047.
- 10 lakh accounts se — $0.0047 \times 10,00,000 = \text{Rs. } 4,700$ chupchaap.
- Agar fraction 0.50 se 0.90 ka ho to — lakhon rupaye.

Kisi bhi account holder ko notice nahi hota — kyunki usko hamesha puri value hi mil rahi thi (round off). Total bekaar me kai lakh rupaye chori — ye Salami Attack hai.

DEFINITION

Salami Attack (Salami Technique) ek cyber crime hai jisme bahut se accounts / files se bahut chhote-chhote amounts is tarah cut kiye jaate hain ki koi notice na kare, lekin total badi amount ban jati hai.

Simple: Har jagah se thoda-thoda — kisi ko pata na chale — lekin total milakar bada gola.

Naam 'Salami' Kyu?

Salami (sausage) ko bahut patle-patle slices me kaata jata hai — har slice kisi ko nazar nahi aata, lekin dheere-dheere poori sausage khatam ho jaati hai. Bilkul isi tarah is attack me chhote-chhote hisse kaate jaate hain — isiliye naam Salami Attack.

FLOWCHART

Bank system (lakhon accounts)



Interest rounding: Rs 100.0047 → 100.00



Fraction (0.0047) attacker ke account me



Har account se chhota loss (notice nahi hota)



Lakhon accounts × chhota amount = Badi chori



Kisi ko pata nahi chalta / trace karna mushkil

Formula (Diagram)

DIAGRAM

Chhota Amount × Bahut Saare Accounts = Badi Chori

Rs. 0.0047 × 10,00,000 accounts = Rs. 4,700 Rs. 0.50 × 10,000,000 accounts = Rs. 50,00,000

(Har account ko loss itna chhota ki wo notice hi na kare)

Salami Technique Ke Aur Examples

- Billing system — har bill par Rs. 1 extra (rounding), 1 lakh bills = Rs. 1 lakh chori.
- Currency exchange — fraction ka paisa (paisa value) wapas na dena, collect karte jaana.
- Payroll / pension — har employee ke salary se Rs. 2-3 katna — kisi ko pata nahi chalta.

Prevention (Bachao Ke Tarike)

- Audit trails — har transaction ka pura record.
- Rounding / fraction ke changes par monitoring.
- Internal controls + dual approval — ek banda akela change na kar sake.
- Regular reconciliation — saare accounts match karna (deposits vs records).
- Fraction data ko bhi log karna.
- Whistle-blower policy — employees bina dar ke report karein.

EXAM NOTES (2 MARKS / VIVA)

Exam Notes (2 Marks):

1. Salami Attack = bahut chhote-chhote amounts, bahut saare accounts se, notice na hote hue — total bada. 2. Naam 'Salami' isiliye — jaise sausage ke patle slices kaatte hain, waise hi chhote hisse kaate jaate hain. 3. Formula: Chhota amount × Lakhon accounts = Badi chori. 4. Ye itna dangerous hai kyunki detection bahut mushkil hota hai — koi bhi ek account owner notice nahi karta.

Viva Q1. Salami attack me har account ko kitna loss hota hai? Ans: Bahut chhota (paisa-fraction) — isiliye notice nahi hota.

Viva Q2. Salami attack ko pakadna mushkil kyu? Ans: Kyunki har jagah amount na ke barabar hai, sirf total bada hota hai — aur rounded values ki wajah se kisi ko difference dikhta nahi.

MEMORY TRICK

Salami = Sausage ki patli-slice. Jaise patle-patle slices se poori sausage khatam — waise hi chhote-chhote cut se poori daulat khatam. Socho: 1 rupaya × 1 crore accounts = 1 crore रुपये — kisi ko pata bhi nahi chala.

APPENDIX D – QUICK REVISION SHEET (UNIT 1)

Sabse Pehle Yaad Rakho — One-Line Definitions | Topic | Ek Line Me

Cyber Security Computers, networks, data ko attacks se protect karne ka process

Cyber Attack System/data ko access, steal, damage karne ka intentional attempt

Cyber Crime Computer/internet ke through ki gayi criminal activity

Email Spoofing | Sender ka address fake karke trusted dikhana

Spamming Bina permission ke bulk unwanted messages bhejna

Internet Time Theft Doosre ka paid internet/bandwidth/time bina permission use karna

Salami Attack Chhote-chhote amounts, bahut saare accounts se, notice na hote hue

Memory Tricks

- Cyber Security = Digital Ghar ki Security.
- Cyber Attack = Digital Chor.
- Cyber Crime = Digital Daku.
- Salami = Sausage ki patli-slice — 1 रुपया × 1 crore accounts = 1 crore रुपये.
- CIA = Confidentiality, Integrity, Availability.
- Active Attack = Badlaav karta hai, Passive Attack = Sirf dekhta hai.

Rapid Fire Viva (10 Questions)

Q | Answer | Cyber security kya protect karti hai? | Computer, network, server, mobile, data | Cyber attack ka sabse common motive? | Paisa (financial gain) | Cyber crime ke 5 elements? | Offender, Tool, Victim, Act, Loss

Cyber crime ki 3 categories? Against Person, Against Property, Against Government

Email spoofing me kya fake hota hai? | Sender ka address | Email spoofing kyu possible hai? | SMTP sender ko verify nahi karta | Spam ka carrier kya hota hai? | Scam / phishing / malware | Internet Time Theft ke 2 types? | Employee Time Theft, Connection Theft

Salami attack me kitna loss per account? Na ke barabar (chhota) — isiliye notice nahi hota

Sabse mushkil cyber crime pakadna? | Salami attack (koi notice nahi karta)

Note: Ye Unit 1 ka complete notes hai. Aage ke chapters (Active/Passive Attack detail, Malware types, Phishing, Virus etc.) same pattern par alag book me banenge.